

# MADHURA JOSHI

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## EDUCATION

### B.Tech, Chemical Engineering | Indian Institute of Technology (IIT), Madras

CGPA: 7.87/10 | 2024 - 2028\*

- Academic achievements:** Secured \*\*99.63 percentile\*\* in JEE-Mains 2024 (AIR 5950); Placed in the top tier of JEE Advanced (AIR 9060) out of 1.5L+ candidates.
- Relevant Coursework:** Programming and Data Structures in Python, Machine Learning Foundations, Computational Techniques, Probability & Statistics, Kinetics & Catalysis, Chemical Engineering Thermodynamics.

## TECHNICAL SKILLS

|                             |                                                                                                                 |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------|
| Programming Languages:      | Python (Expert), SQL, MATLAB, JavaScript, Java                                                                  |
| Libraries & Frameworks:     | PySpark, Scikit-Learn, Pandas, NumPy, Plotly, PyTorch, Matplotlib, Seaborn                                      |
| Tools & Enterprise Systems: | Streamlit, SQLite (LIMS Simulator), Git, GitHub, Microsoft Excel, Power BI                                      |
| Core Competencies:          | Formulation Science Modeling, Property Simulation, AI Agent Workflows, Process Mapping, Multi-Output Regression |

## PROFESSIONAL EXPERIENCE

### Machine Learning Intern | [Company / Research Lab]

May 2025 - June 2025

- R&D; Data Pipelines:** Designed Python data pipelines to clean, preprocess, and structure experimental measurements, streamlining exploratory data analysis (EDA) for new formulations.
- Predictive Modeling:** Trained and evaluated predictive machine learning models in Scikit-Learn to reduce physical lab trial-and-error cycles.
- Data Storytelling:** Visualized multi-variable experimental correlations and key physical trends, communicating findings to senior R&D; team leads.

### Co-Founder & Business Lead | #DotConnect (DotEco Founding Circle)

Dec 2025 - Present

- Process Mapping:** Mapped business workflows and established key performance indicators (KPIs) to track user growth metrics.
- Visual Analytics:** Built interactive business performance tracking dashboards, supporting executive decision-making.

## TECHNICAL PROJECTS

### FormuAI: Formulation Optimizer & Digital Lab Agent (FMCG R&D;)

June 2026

- Formulation Modeling:** Trained a multi-output RandomForestRegressor model on synthetic ingredient ratios (surfactants, co-surfactants, thickeners, actives) to predict critical shampoo properties (viscosity, pH, foam volume, stability), reducing trial-and-error cycles from days to < 10ms.
- AI Lab Copilot Agent:** Developed an NLP search and optimization agent that translates chemist instructions (e.g. 'Show experiments with stability > 300 days') into SQLite SQL commands or simulation parameters.
- LIMS & BI Integration:** Set up a relational SQLite database simulating LIMS/ELN records and built an interactive Streamlit dashboard mapping ingredient correlations, mimicking enterprise Power BI reports.

### LumiFoam: Computer Vision Foam & Emulsion Analyzer

May 2026

- Computer Vision Pipeline:** Built OpenCV image processing pipelines using Hough Circle Transforms to automate droplet size measurements of surfactant emulsions.
- Decay Curve Fitting:** Applied SciPy exponential curve-fitting models to calculate shampoo foam volume decay half-life, automating physical lab measurements.

### Adsorption Modeling in Porous Solids & Heterogeneous Systems

May 2026

- Numerical Solvers:** Formulated a MATLAB mathematical model solving coupled partial differential equations (PDEs) for mass and energy transport in heterogeneous catalyst pellets.

- **Optimization:** Analyzed internal reactant distributions and Thiele modulus across complex geometries to prevent starvation regimes.

**POSITIONS OF RESPONSIBILITY**

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| <b>Club Head - Fine Arts   Sangam, IIT Madras</b>                                                                                                                                                                                                                                                                                                                                     | <b>May 2025 - June 2026</b> |
| <ul style="list-style-type: none"><li>• <b>Cross-Functional Leadership:</b> Governed financial budgeting and led a multi-tiered team of 9 coordinators and a 35-member contingent.</li><li>• <b>Process &amp; Documentation:</b> Drafted operational blueprint reports and financial decks, coordinating event marketing campaigns that increased engagement by <b>20%</b>.</li></ul> |                             |